

TSENTRAL'NYY RUDNIK, Russian Federation

WMO: 296540

Lat: 55.206N

Lon: 87.642E

Elev: 1634

StdP: 13.85

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-21.3	-32.8	0.9	-26.9	-26.9	1.3	-21.0	21.9	20.2	19.6	18.0	2.1	180	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	81.8	63.4	78.3	61.6	75.2	60.4	65.5	76.6	63.9	74.0	62.4	71.8	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	87.9	68.9	60.2	82.6	67.6	58.6	77.9	66.4	31.4	76.6	30.1	74.5	29.0	71.8	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.5	14.3	DB	-31.3	87.6	8.1	2.9	-37.1	89.7	-41.9	91.4	-46.4	93.0	-52.2	95.1	(4)
(5)			WB	-31.5	68.8	8.0	2.0	-37.3	70.3	-41.9	71.5	-46.4	72.6	-52.2	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	5.0	9.5	22.1	34.8	47.2	59.5	62.5	58.9	47.1	34.3	18.4	9.1		
	DBStd	22.63	13.65	11.87	10.93	10.12	9.77	7.67	6.16	6.66	8.56	10.09	12.93	12.31		
	HDD50	6915	1394	1134	865	459	174	16	1	7	154	495	949	1267		
	HDD65	11348	1859	1554	1330	905	556	192	121	210	538	952	1399	1732		
	CDD50	1137	0	0	0	4	86	301	388	283	67	8	0	0		
	CDD65	96	0	0	0	0	3	27	43	22	1	0	0	0		
	CDH74	942	0	0	0	0	66	299	341	211	25	0	0	0		
	CDH80	176	0	0	0	0	13	53	63	44	3	0	0	0		
	CDH80	176	0	0	0	0	13	53	63	44	3	0	0	0		
(14) Wind	WSAvg	5.9	6.7	6.7	7.1	6.4	5.5	4.7	4.1	4.2	5.0	6.3	6.9	7.2		
(15) Precipitation	PrecAvg	38.9	2.2	1.9	2.6	2.4	3.1	3.9	3.7	4.0	3.7	4.3	3.9	3.4		
	PrecMax	52.5	5.0	4.3	11.2	5.4	7.4	7.6	6.3	9.2	7.0	6.7	6.4	11.4		
	PrecMin	30.1	0.3	0.2	0.6	0.4	0.6	1.3	1.1	1.6	0.0	2.1	2.2	1.2		
	PrecStd	6.1	1.1	0.9	2.1	1.1	1.7	1.7	1.4	1.7	1.6	1.2	1.2	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	35.6	52.8	64.3	81.3	85.2	85.9	85.5	77.6	64.8	48.2	34.0		
		MCWB	29.4	30.1	39.7	46.8	59.7	63.4	65.8	65.5	58.0	49.3	40.3	30.5		
	2%	DB	29.0	31.4	45.8	59.2	74.2	80.9	81.2	79.8	70.4	58.7	42.2	30.6		
		MCWB	26.9	27.6	35.5	44.6	55.7	62.2	64.1	63.1	55.1	46.4	37.7	28.7		
	5%	DB	25.1	28.2	40.8	54.6	69.2	77.4	77.6	75.3	65.4	53.9	37.4	27.1		
		MCWB	23.0	25.7	33.0	42.3	52.5	60.4	62.9	61.1	53.3	44.3	33.4	25.3		
	10%	DB	22.1	25.3	36.8	50.2	64.2	73.8	74.3	71.5	60.4	48.5	33.2	24.4		
		MCWB	20.3	23.1	30.7	40.5	50.8	59.1	61.6	59.6	51.2	42.0	31.2	22.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.5	31.9	40.7	49.0	61.6	67.0	67.6	67.9	60.0	51.2	42.0	31.5		
		MCDB	33.3	33.4	51.0	60.5	77.9	79.0	78.8	81.6	73.3	60.3	45.1	33.2		
	2%	WB	27.1	28.0	36.7	46.6	57.4	64.2	65.7	64.7	56.7	48.1	37.7	28.9		
		MCDB	29.0	30.6	43.6	56.3	69.2	75.8	77.0	74.9	67.4	56.0	41.9	30.4		
	5%	WB	23.1	25.7	33.6	43.7	54.8	62.5	64.4	62.8	54.6	45.2	34.1	25.5		
		MCDB	24.6	27.9	39.7	52.7	65.2	73.1	74.4	72.2	63.1	52.7	37.1	27.0		
	10%	WB	20.3	23.2	31.3	41.0	52.1	60.8	62.9	60.9	52.4	42.4	31.4	22.9		
		MCDB	22.1	25.5	36.1	49.1	62.2	71.0	71.8	69.9	59.0	48.0	33.1	24.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	14.6	16.2	16.5	18.5	18.2	16.4	17.0	15.2	12.1	10.9	11.3		
		MCDBR	13.7	15.8	22.3	22.8	27.2	24.7	22.1	23.7	24.5	20.8	12.4	12.1		
		MCWBR	12.0	13.1	14.2	12.8	13.4	10.2	9.2	10.2	11.7	11.4	8.5	10.8		
		MCDBR	12.4	14.2	19.3	20.5	22.8	20.3	18.6	19.8	20.5	18.4	12.1	11.3		
	5% WB	MCWBR	11.4	12.3	13.0	12.6	12.6	9.8	8.9	9.9	10.9	10.8	8.8	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.333	0.383	0.412	0.418	0.410	0.433	0.415	0.378	0.363	0.331	0.299		
	taud		1.976	1.932	1.854	1.954	2.144	2.277	2.218	2.239	2.282	2.160	1.971	1.918		
	Ebn at Noon		183	220	238	253	261	266	256	252	245	212	162	144		
	Edh at Noon		23	35	48	50	44	39	41	38	32	28	23	20		
(44) All-Sky Solar Radiation	RadAvg		277	591	1012	1369	1584	1851	1674	1373	908	465	230	171		
	RadStd		32	47	97	94	192	163	119	109	114	52	39	29		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page